Battery Management System

AUTOSAR High-Level Design Document

Document Version: 2.3

Classification: Internal Engineering Use Only

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1. System Overview

This document describes the High-Level Design (HLD) of the Battery Management System (BMS) implemented using AUTOSAR methodology. The system is composed of three primary Software Components (SWCs): BatteryManagement, ThermalManagement, and PowerControl. Each SWC is an AtomicSWC running on the AUTOSAR Runtime Environment (RTE). Communication between SWCs is mediated through the RTE using defined PortInterfaces.

The BMS ECU runs on a 32-bit microcontroller. The BSW (Basic Software) layer provides services including OS scheduling, NvM, COM, and diagnostic modules. SWC communication follows the Sender-Receiver (S/R) and Client-Server (C/S) paradigms as defined in the AUTOSAR standard.

2. Software Component Descriptions

2.1 BatteryManagement SWC

The BatteryManagement SWC is responsible for monitoring cell voltages, state of charge estimation, and fault detection. It is an AtomicSWC with one InternalBehavior containing three Runnables: Runnable_BattInit (initialisation), Runnable_BattMonitor (10 ms cyclic), and Runnable_BattFault (event-triggered).

Ports:

Port Name	Direction	Interface	Description
BattVoltage_Prov	P-Port	BattVoltage_If	Provides cell voltage data
BattSOC_Prov	P-Port	BattSOC_If	Provides state-of-charge
BattTemp_Req	R-Port	TempSensor_If	Requires temperature reading
BattFault_Prov	P-Port	FaultStatus_If	Provides fault status flag

2.2 ThermalManagement SWC

The ThermalManagement SWC reads temperature sensors and provides thermal data to other SWCs. It also controls cooling fan output based on temperature thresholds. It is an AtomicSWC with one Runnable: Runnable_ThermalCycle (20 ms cyclic).

Ports:

Port Name	Direction	Interface	Description
TempSensor_Prov	P-Port	TempSensor_If	Provides temperature readings
CoolingCmd_Prov	P-Port	CoolingCmd_If	Provides cooling fan command
BattFault_Req	R-Port	FaultStatus_If	Requires battery fault flag

2.3 PowerControl SWC

The PowerControl SWC manages the main contactor relay and power distribution logic. It decides when to enable or disable battery output based on state of charge, fault conditions, and thermal status. It is an AtomicSWC with Runnables: Runnable_PowerInit and Runnable_PowerControl (5 ms cyclic).

Ports:

Port Name	Direction	Interface	Description
BattSOC_Req	R-Port	BattSOC_If	Requires state-of-charge
BattVoltage_Req	R-Port	BattVoltage_If	Requires cell voltage data
CoolingCmd_Req	R-Port	CoolingCmd_If	Requires cooling fan command
RelayCmd_Prov	P-Port	RelayCmd_If	Provides relay enable/disable
BattFault_Prov	P-Port	FaultStatus_If	Note: should be R-Port (intentional inconsistency)

3. Port and Interface Definitions

All PortInterfaces used in this system follow the AUTOSAR SenderReceiver or ClientServer paradigm. The table below lists all defined interfaces.

Interface Name	Type	DataElements / Operations	Units
BattVoltage_If	SenderReceiver	CellVoltage_Val	mV
BattSOC_If	SenderReceiver	SOC_Val	%
TempSensor_If	SenderReceiver	Temperature_Val	°C
FaultStatus_If	SenderReceiver	FaultActive_Bool	boolean
CoolingCmd_If	SenderReceiver	FanDutyCycle_Val	%
RelayCmd_If	SenderReceiver	RelayEnable_Bool	boolean

4. Signal List

The following signals are exchanged between SWCs via the AUTOSAR RTE. All signals are transmitted using the implicit S/R communication mechanism unless stated otherwise.

Signal Name	Producer SWC	Consumer SWC	Update Rate
CellVoltage_Sig	BatteryManagement	PowerControl	10 ms
SOC_Sig	BatteryManagement	PowerControl	10 ms
Temperature_Sig	ThermalManagement	BatteryManagement	20 ms
FaultActive_Sig	BatteryManagement	ThermalManagement	Event
FanDutyCycle_Sig	ThermalManagement	PowerControl	20 ms
RelayEnable_Sig	PowerControl	BSW_IoHwAb	5 ms
OrphanSignal_Sig	BatteryManagement	—	100 ms

5. Integration Notes

The following notes apply during software integration on the target ECU:

• The BatteryManagement SWC must be initialised before the PowerControl SWC. Ensure Runnable_BattInit completes before the first invocation of Runnable_PowerControl.

• The ThermalManagement SWC reads temperature via a hardware abstraction layer (IoHwAb). Ensure the IoHwAb SWC is mapped to the correct ADC channel for the temperature sensor.

• The RTE generation tool must be configured with the correct task mapping for all cyclic runnables. Runnable_BattMonitor (10 ms) and Runnable_PowerControl (5 ms) share OS Task T2.

• The FaultStatus_If interface must be configured with InitValue = FALSE in the AUTOSAR configuration tool. Failure to set this may result in a spurious fault at startup.

• Signal OrphanSignal_Sig is defined in the signal list but currently has no consumer SWC assigned. This should be resolved before final integration.